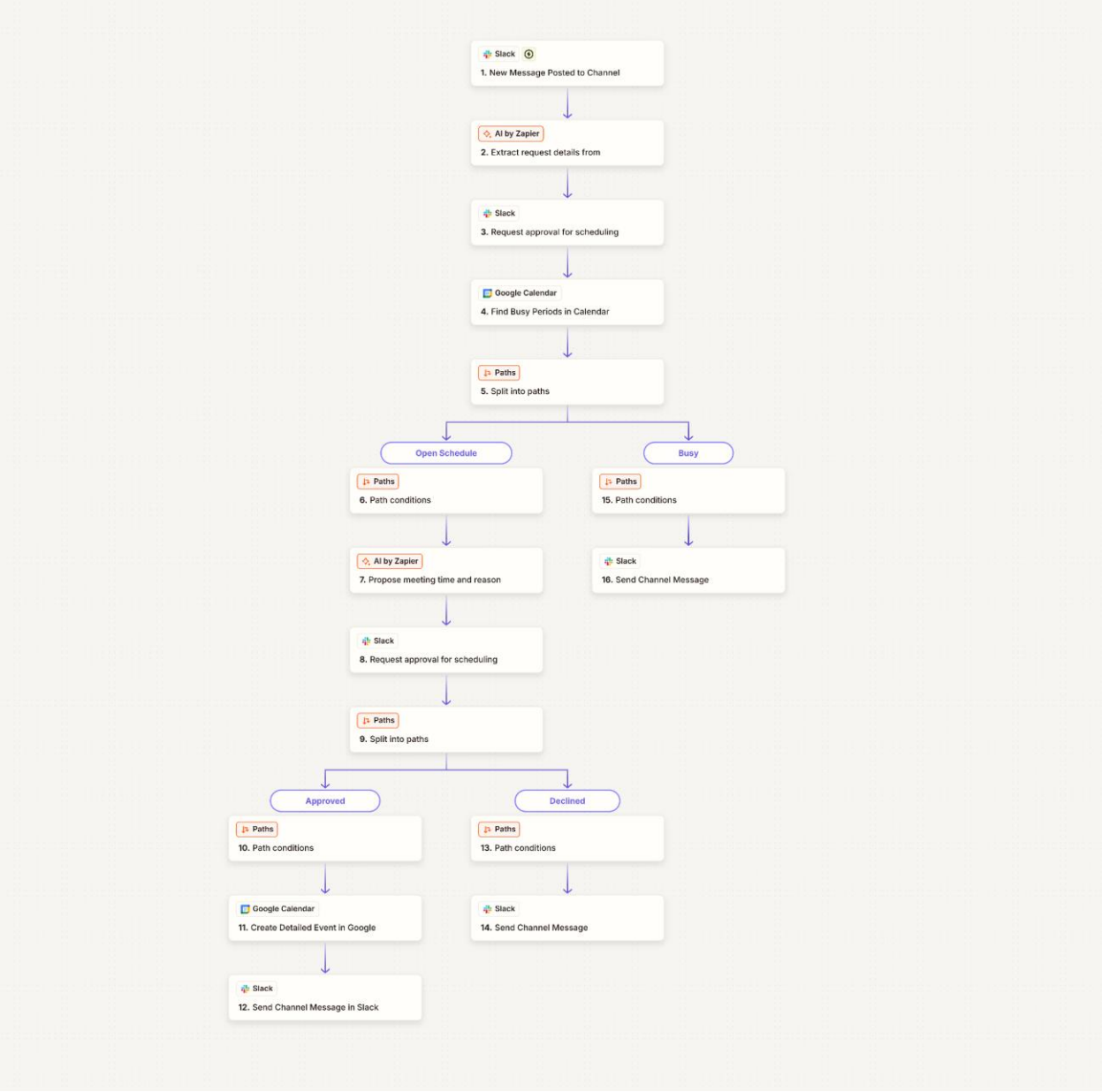
SLACK TO GOOGLE CALENDAR AI SCHEDULING AUTOMATION

Automation VA Portfolio Case Study



Workflow overview from the project documentation

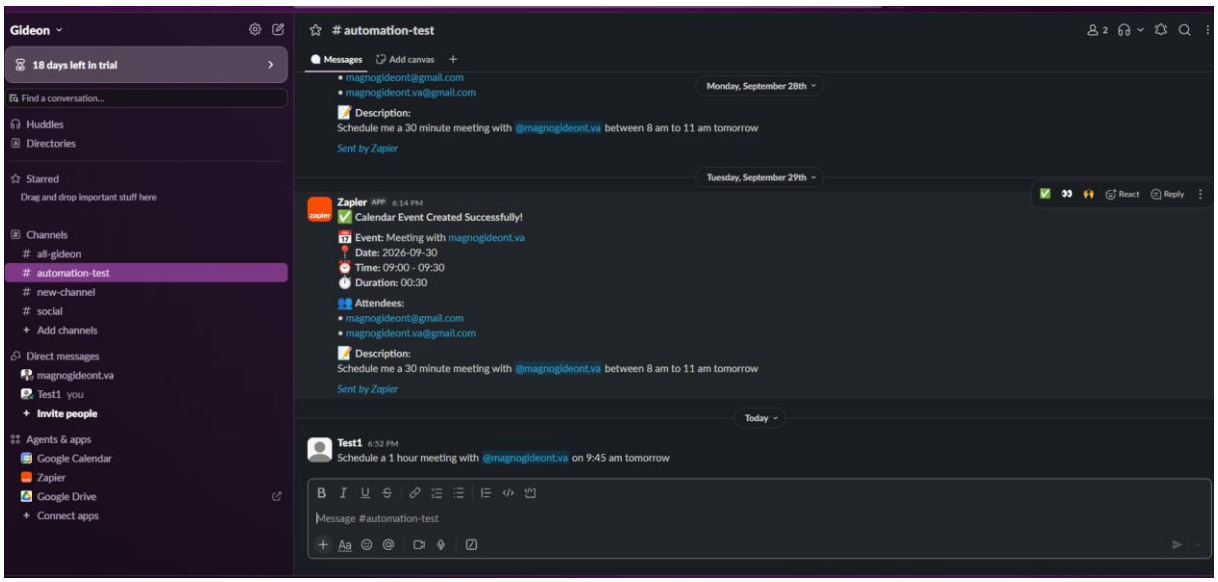
Role: Automation VA / Workflow Builder
Project Type: Personal Automation Portfolio Project
Tools: Zapier, Slack, AI by Zapier, Google Calendar
Status: Working portfolio demonstration

Project Summary

Built a multi-step scheduling automation that receives meeting requests through Slack, extracts scheduling details with AI, checks Google Calendar availability, routes the request through conditional paths, and requires administrator approval before creating the calendar event.

1. The Problem

A scheduling request can require several manual steps before a calendar event is created. The request must be interpreted, the requested period checked against the calendar, a suitable time proposed, and the final event approved and created.



Example request intake from the project trial

2. The Goal

The workflow was designed to:

- Receive a natural-language scheduling request in Slack.
- Extract the scheduling information needed by the workflow.
- Check the requested period against Google Calendar.
- Separate open schedules from busy schedules.
- Propose a meeting time when the requested period is available.
- Send the proposed schedule for administrator approval.
- Create the calendar event after approval.
- Handle declined and busy requests with Slack notifications.

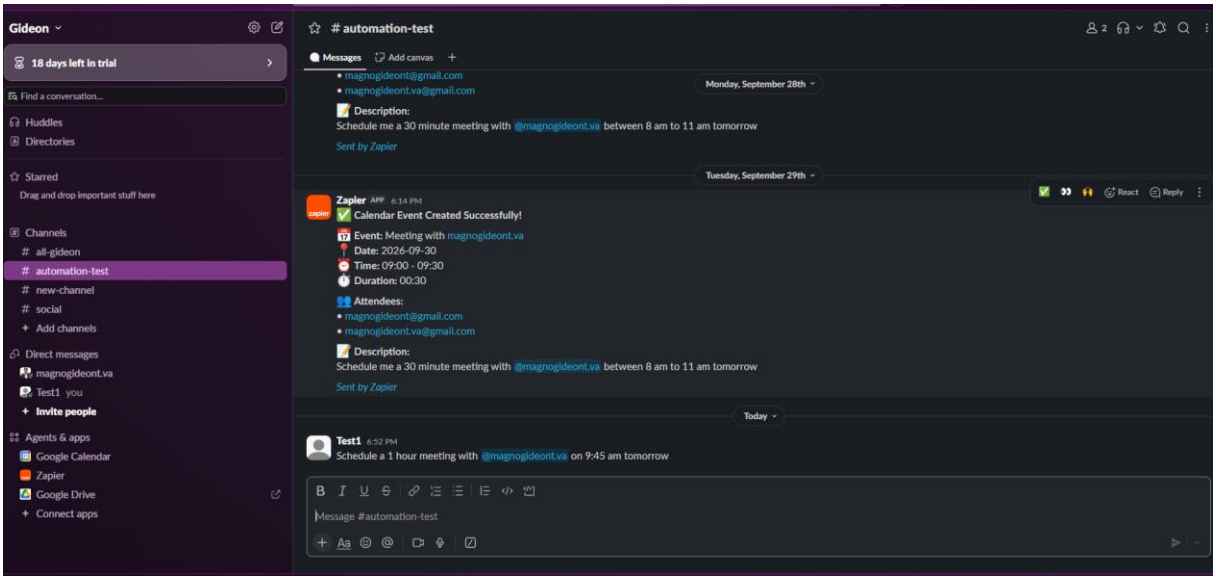
3. Solution Architecture

The automation connects four main components: Slack for request intake and approval, AI by Zapier for request interpretation and scheduling proposals, Google Calendar for availability and event creation, and Zapier for workflow orchestration and conditional routing.

4. Request Intake and AI Extraction

4.1 Slack Request

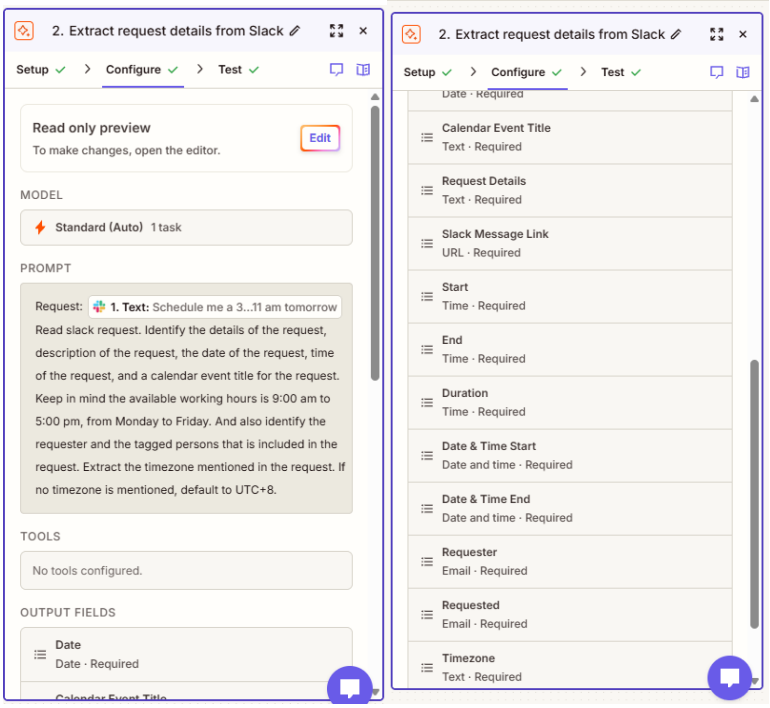
The process begins when a requester posts a scheduling request in the designated Slack channel.



Slack request used in the documented workflow

4.2 AI Request Extraction

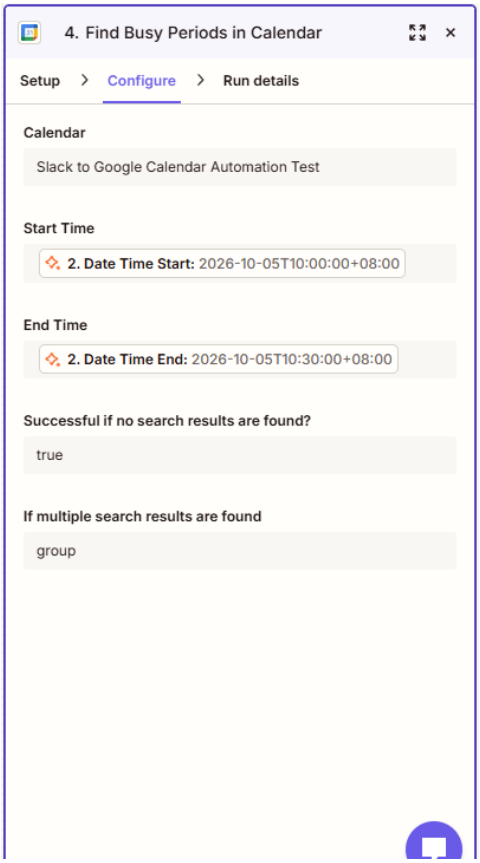
AI by Zapier converts the natural-language request into structured scheduling information used by later workflow steps.



AI by Zapier request extraction.

5. Calendar Availability and Conditional Routing

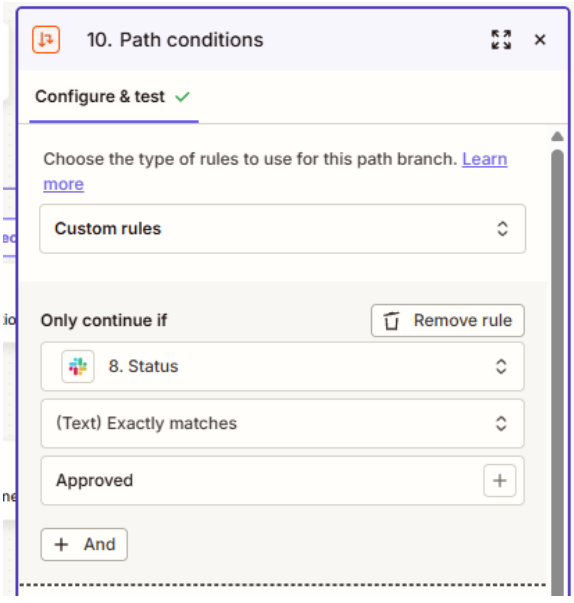
Google Calendar checks the requested period against existing busy periods. The workflow then routes the request according to the availability result.



Using Find Busy Periods to look for availability of requested time.

Open Schedule

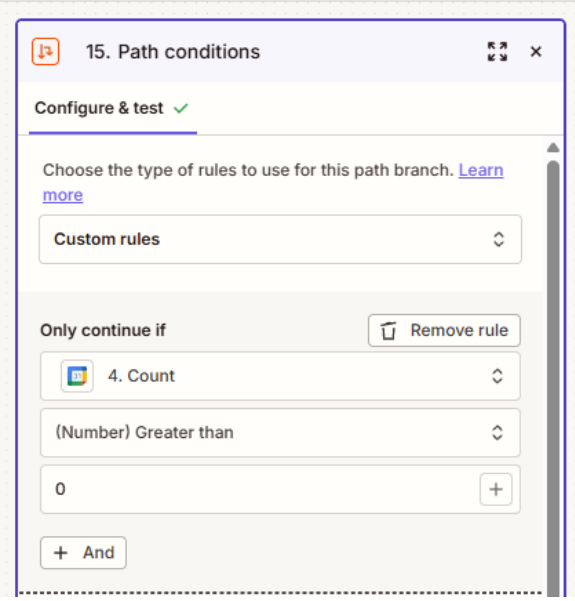
When the requested period is open, the workflow continues toward an AI-generated scheduling proposal.



Open schedule path configuration

Busy Schedule

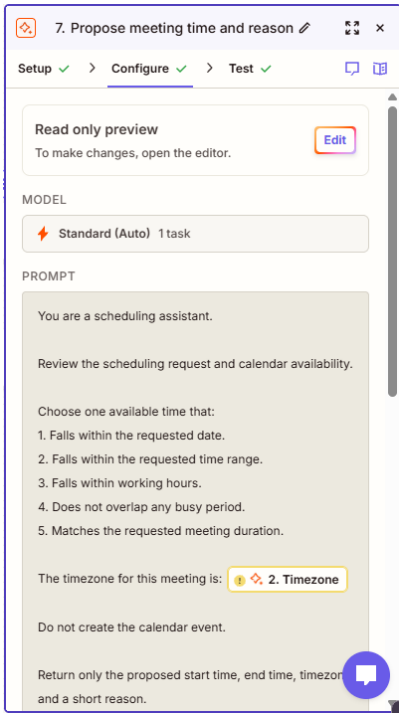
When the requested period overlaps with an existing schedule, the workflow follows the busy path and asks the requester to provide a different time.



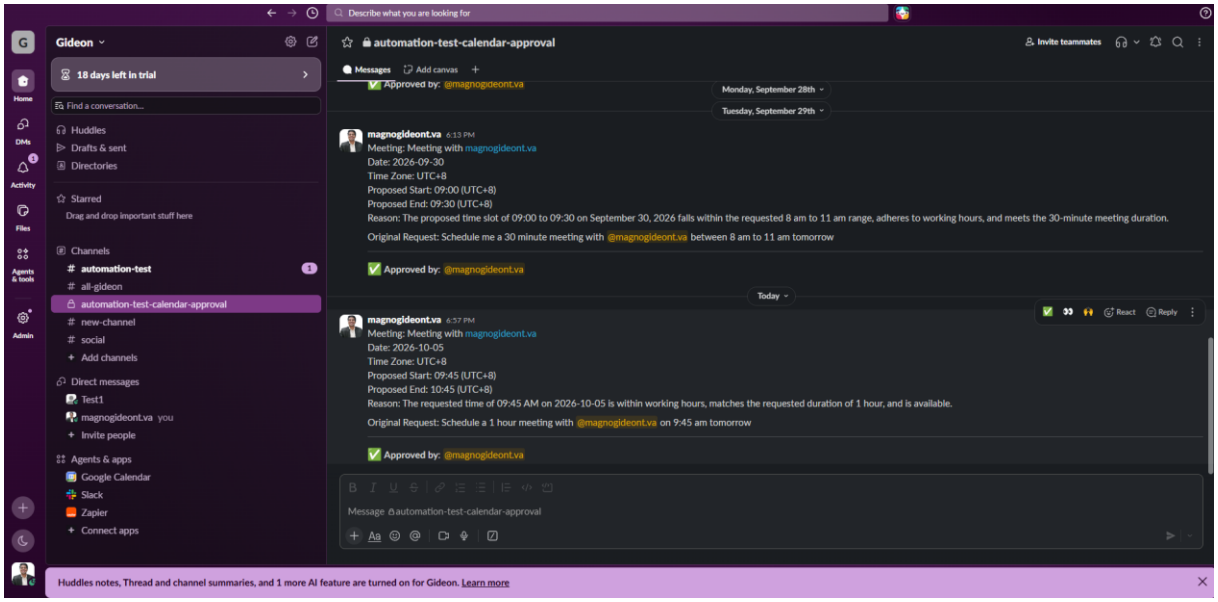
Busy schedule path configuration

6. AI Scheduling Proposal and Approval

For an open schedule, AI by Zapier proposes a meeting time based on the request and availability information. The proposal then goes to Slack for administrator approval.



AI by Zapier reviewing the request and calendar availability.



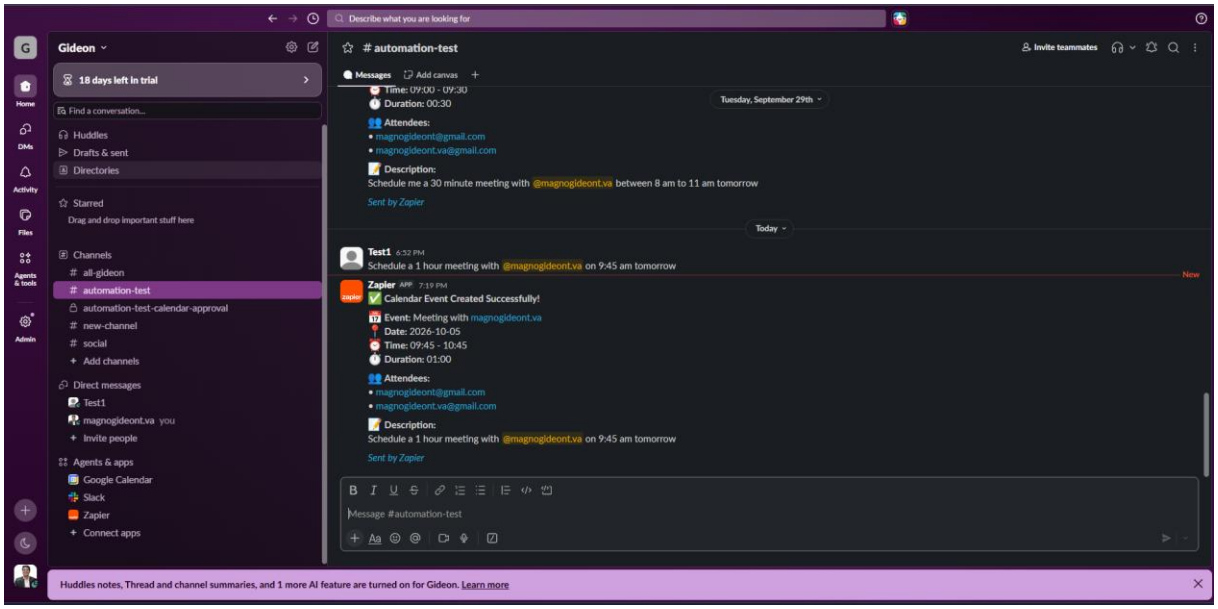
Administrator approval request

Approval Control

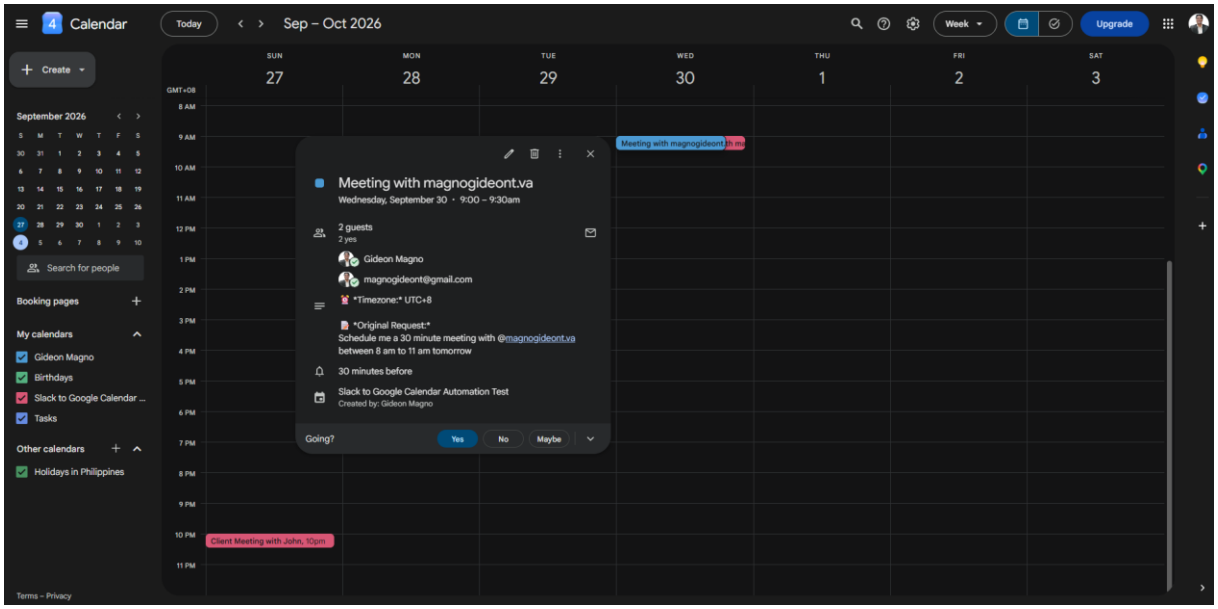
The approval step keeps a human decision between the AI proposal and calendar event creation. This prevents the workflow from creating the event before an administrator approves the request.

7. Automation Trial: Test 1, Approved Event

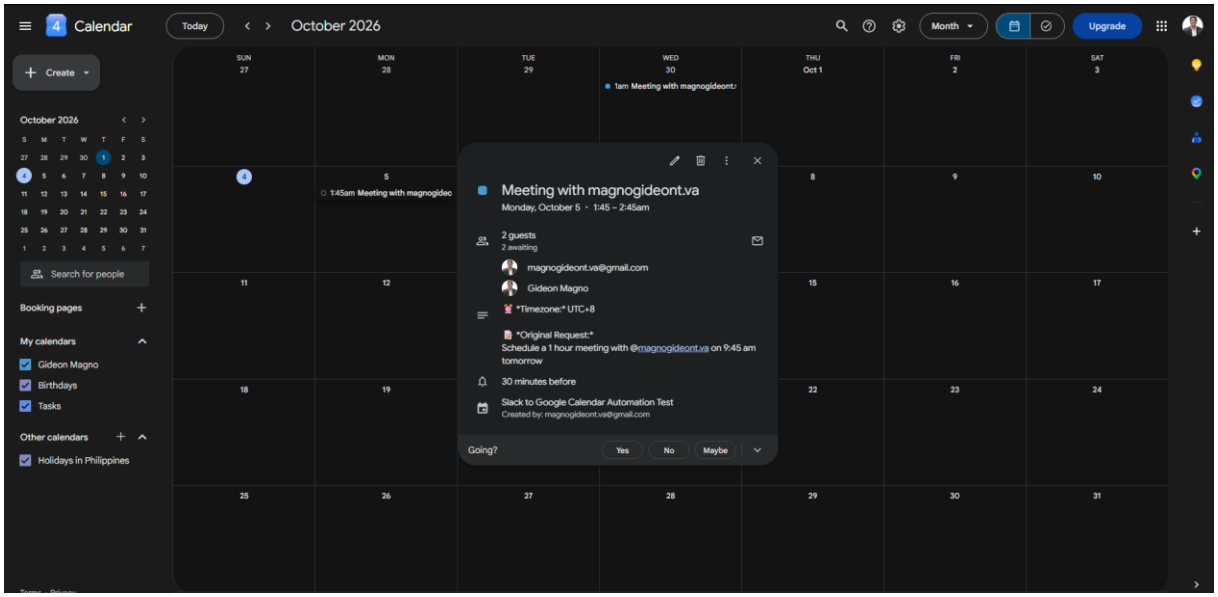
The first documented trial follows an approved scheduling request. The requester submits a meeting request, the workflow presents the request details, and the request proceeds to administrator approval. After approval, the schedule is created and the attendees receive the calendar invitation and email described in the source documentation.



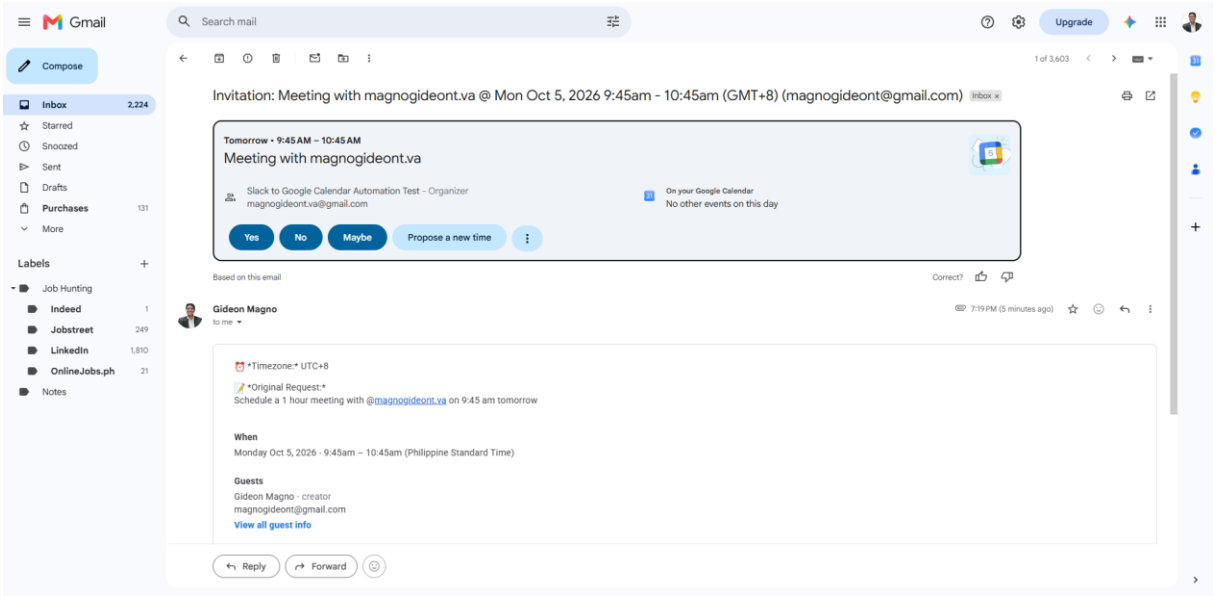
Trial 1: scheduling request and workflow activity



Trial 1: calendar and workflow evidence



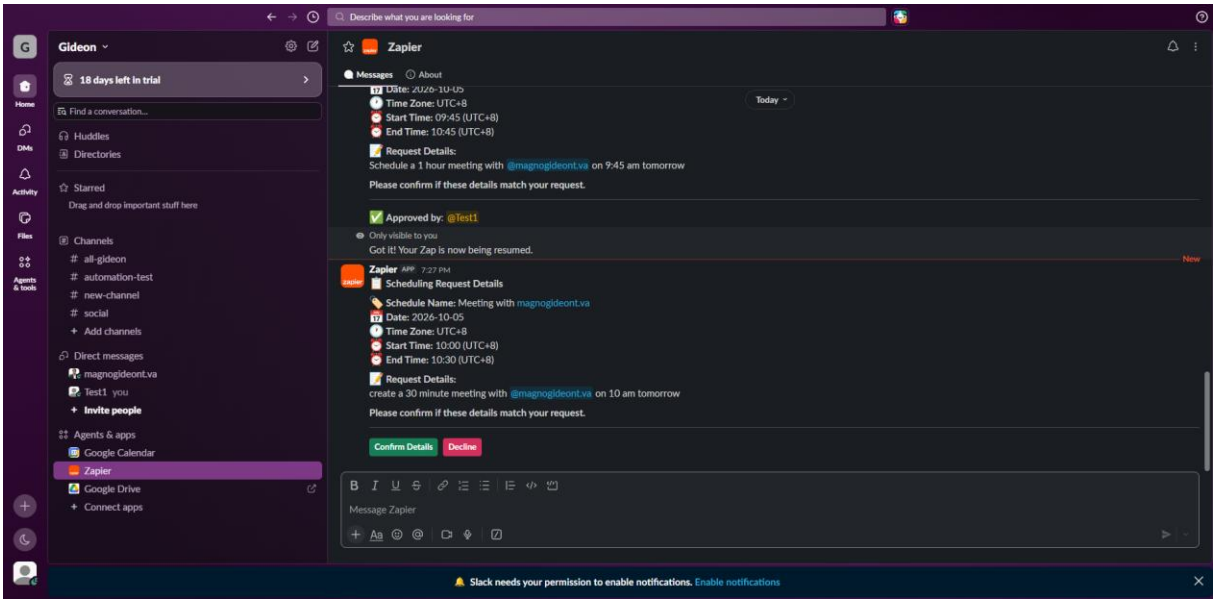
Trial 1: created schedule evidence



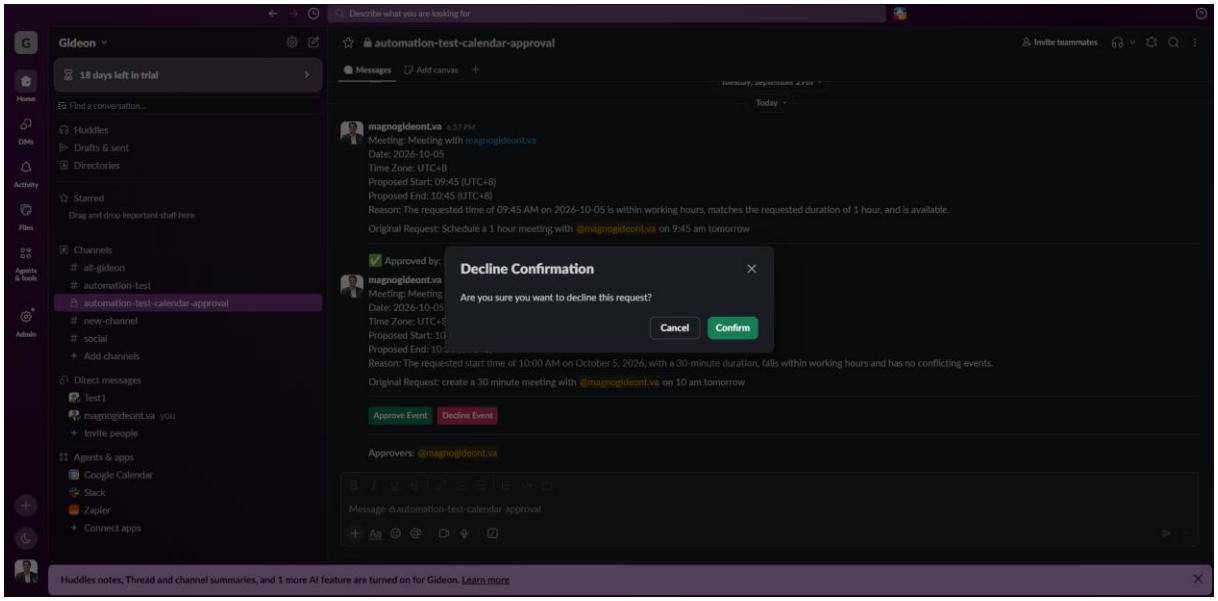
Trial 1: notification or calendar invitation evidence

8. Automation Trial: Test 2, Declined by Administrator

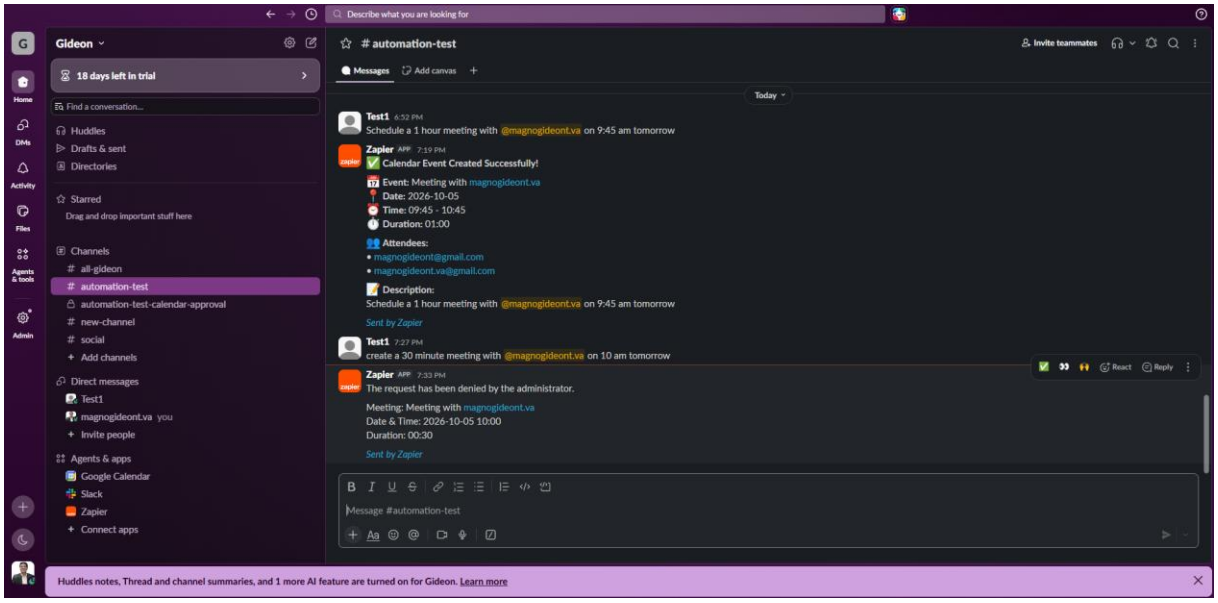
The second documented trial follows a request that reaches the administrator approval stage and is declined. The workflow then sends a notification that the schedule has been declined.



Trial 2: confirmation of the details of the request by the user.



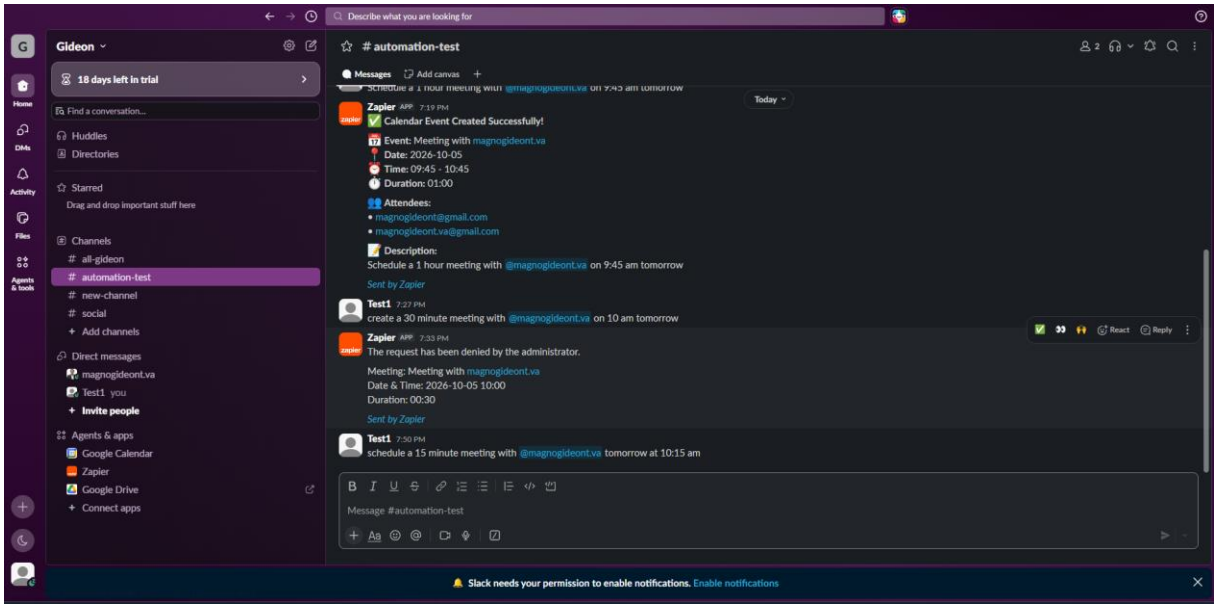
Trial 2: administrator declined the request.



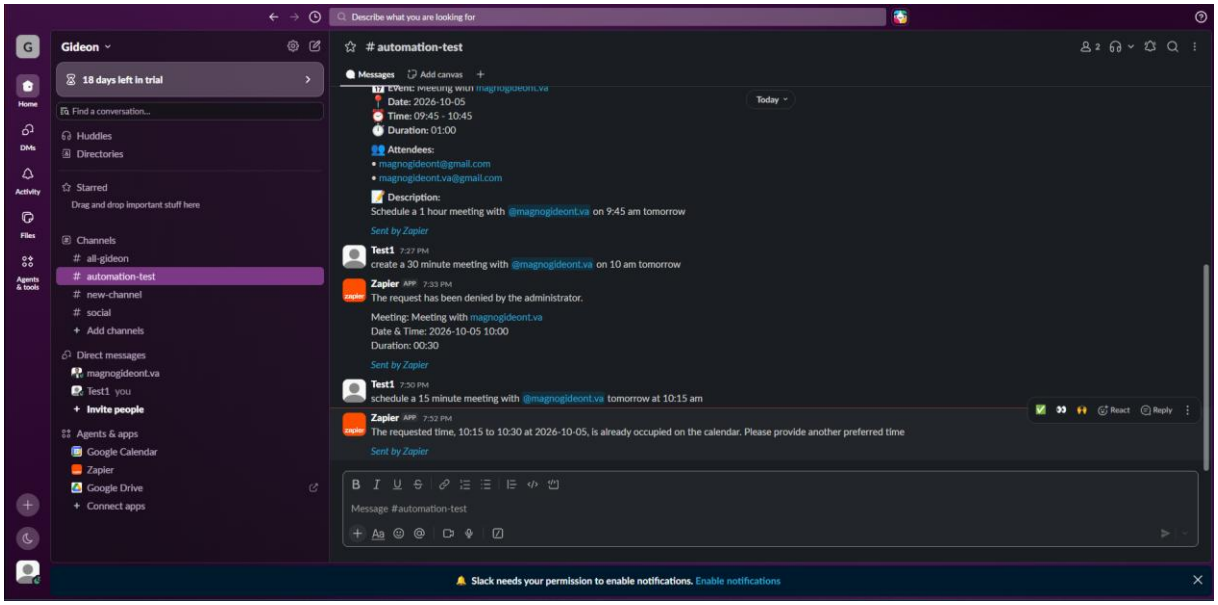
Trial 2: declined status notification

9. Automation Trial: Test 3, Busy Schedule

The third documented trial uses a request that overlaps with an earlier approved schedule. The workflow identifies the conflict and automatically declines the request while asking the user to provide a different time.



Trial 3: user scheduled a meeting on an occupied time.



Trial 3: busy schedule response

10. Automation Logic

Condition	Workflow Response
Schedule is open	Continue to the AI scheduling proposal and approval process.
Schedule is busy	Automatically decline the request and ask for a different time.
Administrator approves	Create the schedule and send the documented confirmation/invitation.
Administrator declines	Send a Slack notification that the schedule was declined.

11. What I Built

- Slack-based scheduling request intake.
- AI-assisted scheduling information extraction.
- Google Calendar availability checking.
- Conditional routing for open and busy schedules.
- AI-assisted meeting-time proposal.
- Slack-based administrator approval.
- Approved and declined workflow branches.
- Busy-schedule handling.
- Google Calendar event creation after approval.
- Slack status notifications.

12. Skills Demonstrated

- Zapier workflow design
- Conditional paths and decision logic
- Slack automation
- Google Calendar automation
- AI-assisted workflow design
- Structured data extraction
- Human approval controls
- Workflow testing
- Exception handling
- Process documentation

13. Potential Business Applications

The workflow can serve as a starting point for scheduling processes used by executive assistants, agencies, consultants, coaches, real estate teams, and small businesses. These are potential use cases for the project, not current client deployments.

14. Current Limitations

- This is a personal portfolio demonstration, not a live client deployment.
- More advanced timezone handling may be needed for multi-region scheduling.
- Recurring meetings may require additional workflow logic.
- Ambiguous requests may require additional validation.
- Attendee information needs reliable contact data before it can be added to the final event.

15. Future Improvements

- Generate alternative time options when the requested period is busy.
- Add attendee lookup and validation.
- Add duplicate-request detection.
- Support recurring meeting requests.
- Support multiple calendars or calendar owners.

16. Portfolio Takeaway

This project demonstrates practical workflow automation using Zapier, Slack, AI by Zapier, and Google Calendar. The workflow is not limited to a successful path. It also handles busy schedules and administrator decisions, showing how conditional logic and approval controls can be combined in an automation.